BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLYPROGRESSREPORT (18/05/2026 – 24/05/2026)

WEEK 1

PROJECT NAME	RADIO PROPAGATION ANALYSIS AND PATH PREDICTION

PROJECT DESCRIPTION : Project focuses on simulating antenna beam propagation and predicting communication coverage.

GROUP MEMBER : GARIMA CHOUDHARY, MANASH PRASAR, ARYAN KUMAR

GROUP ID : 2026G265

GROUP GUIDE : Dr. ABDUL JHUMMARWALA

GROUP COORDINATOR : GARIMA CHOUDHARY

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COLLEGE NAME : GATI SHAKTI VISHWAVIDYALAYA

18/05/2026	• Submitted the internship joining form and completed the documentation process • Met the project instructor, received the project title and initial instructions to run the project
19/05/2026	• Started studying antenna-array radiation patterns and basic GNU Octave environment (Manash prasar) • Learnt basics of beam patterns and signal propagation concepts (Garima choudhary) • Explored basic concepts of RF propagation (Aryan kumar)
20/05/2026	• Studied gain and dB representation in antenna-radiation patterns (Manash prasar) • Generated initial radiation-pattern plots in GNU Octave and studied beam directions (Garima choudhary) • Learnt basics of Friis transmission equation (Aryan kumar)
21/05/2026	• Studied non-uniform antenna-array excitation and side-lobe behavior (Manash prasar). • Compared basic radiation-pattern variations for different beam shapes (Garima choudhary) • Explored introductory RF link-budget concepts (Aryan kumar)
22/05/2026	• Prepared beam-pattern observations and visualization notes (Manash prasar) • Implemented basic propagation-distance estimation using GNU Octave simulations (Garima choudhary) • Explored basic features of propagation-analysis tools (Aryan kumar)
23/05/2026	Holiday (4th Saturday)
24/05/2026	Holiday (Sunday)

Next Week Plan	We are planning to study non-uniform antenna-array behavior in more detail and implement improved radiation-pattern simulations using GNU Octave. We will also explore antenna-simulation and propagation-analysis tools for better path-prediction understanding

REFERENCE:

https://octave.org/doc/interpreter/

https://www.antenna-theory.com/

https://www.qsl.net/4nec2/

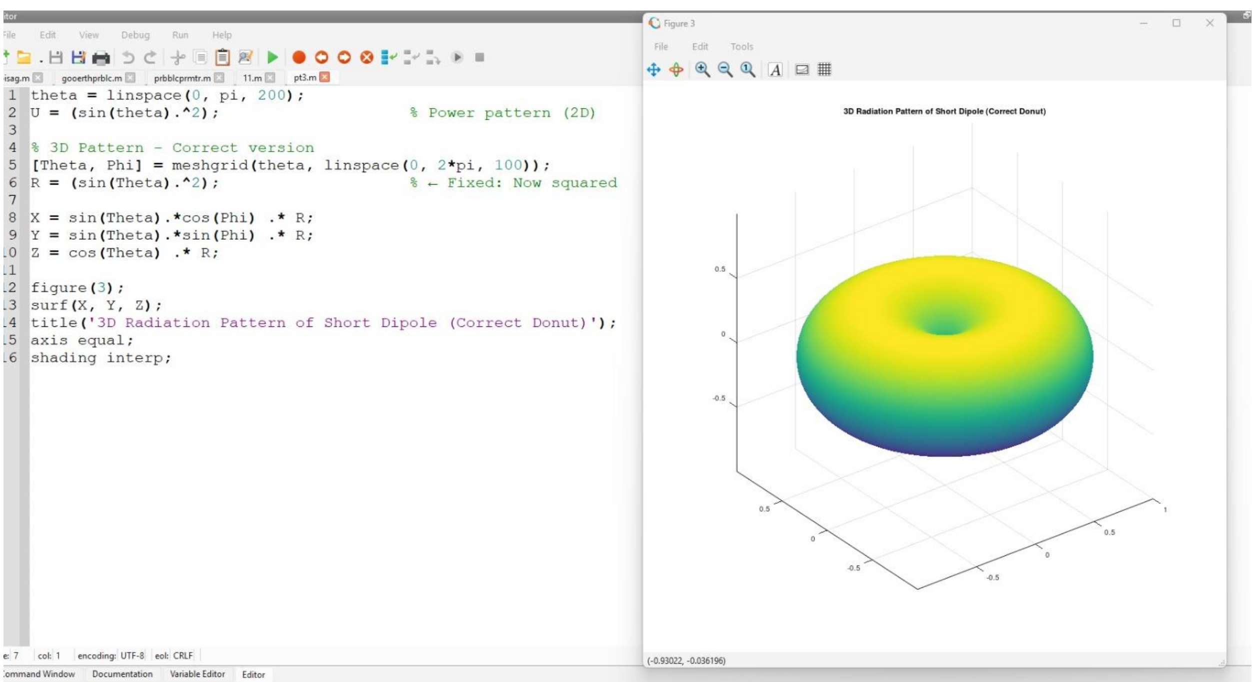
https://www.everythingrf.com/rf-calculators/link-budget-calculator

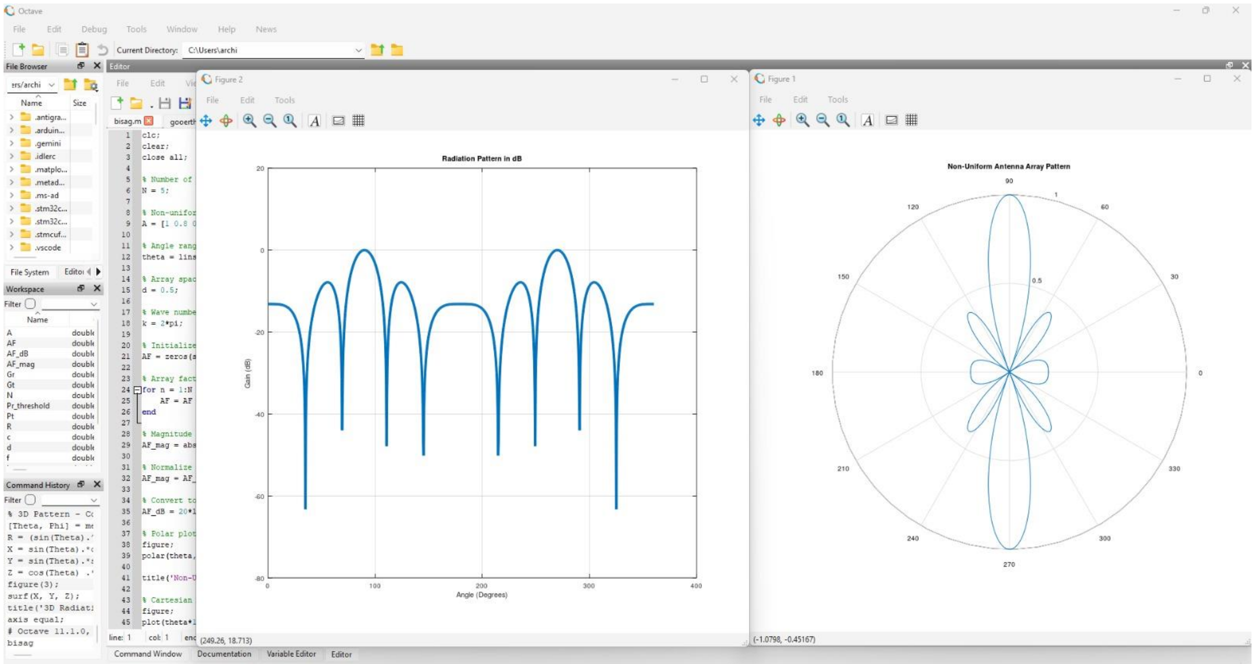
https://sandcastle.cesium.com/

https://www.qsl.net/kd2bd/splat.html

https://www.youtube.com/watch?v=ZaXm6wau-jc

SCREEN SHOTS:




Octave
Figure 2
Radiation Pattern in dB
Gain (dB)
Angle (Degrees)
Figure 1
Non-Uniform Antenna Array Pattern

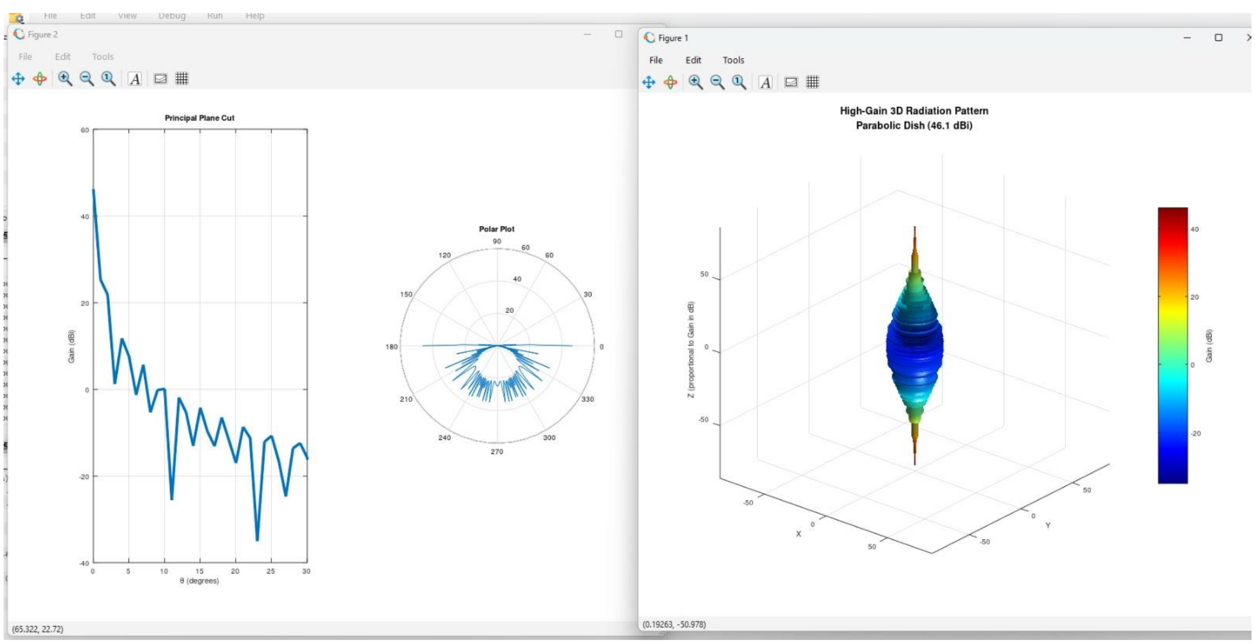


Figure 2
Principal Plane Cut
Gain (dBi)
θ (degrees)
Polar Plot
Figure 1
High-Gain 3D Radiation Pattern
Parabolic Dish (46.1 dBi)
Z (proportional to Gain in dB)
Gain (dBi)